colleague Ayyappan Pillai, an IITian specialising in combustion, I had ascertained that the Viking engine had a low upper stability margin. The engine is designed to have an operating pressure of 52.6 bars (approximately 750 lbs per square inch). The operating pressure of a rocket engine is determined by several factors, primarily by the quantity of fuel and oxidiser entering the combustion chamber. We found out that the engine would become unstable due to vibration when the pressure goes up by about two bars. If, for various reasons the pressure goes up beyond the margin, the engine would blast. A system like this one should have ideally had a minimum upper stability margin of five bars.

How did we find this out? Normally during development, specifications given by a team are taken for granted. Here, we, as part of our learning rigour, never accepted weights and measurements given on paper—we did our own assessment. So, looking at the given operating pressure of 52.6 bars, I was curious what the stability margin was. This, because I wanted to upgrade the engine we were co-producing, once back in India to carry heavier payloads than those planned by Viking. The man to go to was Ayyappan Pillai, our instability expert. Pillai ran a computer programme to tell me that the upper stability margin was low. That is, the engine would start vibrating a couple of bars above the operating pressure, indicating instability. An unstable engine would send the rocket away from the desired trajectory and, worse, trigger an explosion.

I knew Pillai had just handed me a trump card, and I played it. During a casual conversation with Dorville, I let it slip that I felt the stability margin was low. Dorville didn't take it seriously, but he was curious.

'Why do you say that?' he said.

'I get a feeling it is low. And yes, I have my reasons to feel so,' I said.

Dorville neither agreed nor disagreed, but the game did work. Dorville must have reported the information to his senior colleagues. And soon there was a discussion. When asked if I could substantiate my suspicion over Viking's stability, I suggested we run a test to ascertain it. They agreed.

Pillai had done a computer analysis and had employed Professor Crocco's theorem of instability to confirm Pillai's findings.